Name ____ Date ___________ Period _____

Parent Functions II: x^3 and $\sqrt[3]{x}$ - Practice

Algebra II Honors | Unit 1 | Lesson 5 Practice | TEK A2.2.A

Objective: I can prove x^3 and $\sqrt[3]{x}$ are odd algebraically, and use average rate of change to describe how they grow.

Section 1 — Recall

1. Evaluate $f(x) = x^3$ and $g(x) = \sqrt[3]{x}$ so they show they undo each other: $f(3) = ?$ then $g(27) = ?$

$f(3)$ = __________ $g(27)$ = __________

2. Test $h(x) = 2x$ for odd/even using $h(-x)$. Is it odd, even, or neither?

$h(-x)$ = ____________________ ____________

Section 2 — Apply

3. Prove algebraically that $j(x) = 5x^3$ is odd.

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$j(-x)$ = _________________________________

4. Compute the average rate of change of $f(x) = x^3$ on [0, 2] and on [2, 4]. What does the change tell you about how the cubic grows?

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On [0,2]: __________ On [2,4]: __________ ______________________________

Section 3 — Challenge

5. Solve $x^3 = -125$ and $x^2 = 125$ (round to the nearest hundredth if needed), and count the real solutions for each.

$x^3 = -125$: ________________________ $x^2 = 125$: ______________________________

6. Can an odd function ever have a y-intercept other than (0, 0), or be undefined at x = 0? Investigate using $f(-x) = -f(x)$ at x = 0.

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Section 4 — Explain It

7. A classmate claims that because x^3 and $\sqrt[3]{x}$ are both odd and share identical domain, range, and symmetry, they must be the exact same function. Explain why the classmate is wrong, and identify what piece of information would prove they are different.

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